Spoken Tutorial – Bias in AI

Title: Bias in AI

Module	Ethics, Responsibility, and the Future
Episode	Bias in AI
Learning Objective	At the end of this tutorial learner will be able to 1. Understand how human bias enters **AI** training data. 2. Identify how **AI** models learn and amplify existing patterns. 3. Recognize the importance of identifying and reducing bias.
Approx. Duration	2-3 min
Outline	• Bias in **AI**: Where does it start? • **AI** Learning and Amplifying Patterns • Example: Gender Bias in Outputs • Identifying and Reducing Bias
Meta Tags	AI bias, machine learning ethics, data bias, ethical AI, responsible AI, gender bias, AI training data
Pre-requisite Tutorial	You should have a basic understanding of what **AI** is. Familiarity with machine learning concepts helps. No advanced technical knowledge is needed. We will explain all new terms simply.

Script

Visual Cue	Narration
	Welcome to Module 5. We will explore Ethics, Responsibility, and the Future of **AI**. Our episode today focuses on Bias in **AI**. This is a very important topic.
	By the end of this episode, you will understand how human bias enters **AI** training data. You will identify how **AI** models learn and amplify existing patterns. You will also recognize the importance of identifying and reducing bias.
	You will need a device with internet access. A stable connection will help you. You should have a quiet space to learn. Please use headphones for better audio.
	You should have a basic understanding of what **AI** is. Familiarity with machine learning concepts helps. No advanced technical knowledge is needed. We will explain all new terms simply.

	Can an **AI** be fair if its data is not fair? **AI** learns from the data we give it. This data comes from the real world. The real world has human biases. These biases enter the training data. Let us see how this happens. Think of a child learning from old books. These books might show only men in certain jobs. The child learns this pattern. Similarly, **AI** learns from historical data. You saw how data shapes learning. Did the **AI** create this bias itself? Now, we will explore how models use this data.
	How does an **AI** make bias bigger? **AI** models look for patterns in data. They find connections between things. If a pattern is biased, the **AI** will learn it. It then uses this pattern to make decisions. Let us look at a common example. Imagine an **AI** trained on job applications. Historically, more men were hired for engineering roles. The **AI** sees this pattern. When new applications come, it might favor male candidates. It amplifies the existing bias. You saw how patterns are learned. Is the **AI** being unfair deliberately? Next, we will see a specific output example.
	What happens when an **AI** creates images or text? If the training data has gender bias, the **AI** output shows it. For instance, asking for a 'doctor' might only show male images. Asking for a 'nurse' might only show female images. Let us view a clear scenario. We ask an **AI** image generator: 'Show me a CEO.' Often, the **AI** generates images of men. This is because most historical data shows male CEOs. The **AI** simply reflects this. You saw the output reflecting bias. Does this output seem fair to everyone? Now, we understand the problem. What can we do?

	Why is it important to fix **AI** bias? Biased **AI** can lead to unfair decisions. It can harm people and groups. We must ensure **AI** treats everyone fairly. Identifying and reducing bias is crucial. Let us think about the steps. First, we can check our training data for balance. Next, we can use diverse datasets. Finally, we can also test **AI** outputs for bias. This helps us to correct it. You learned about checking for bias. How can we make **AI** more inclusive? We have covered the core ideas. Let us summarize our learning.
	We learned that human biases enter **AI** data. **AI** models learn and amplify these patterns. We saw how gender bias appears in **AI** outputs. Identifying and reducing bias is very important for fair **AI**.
	Think of an **AI** tool you use daily. How might bias affect its outputs? Write down one example. Share your thoughts in the comments.
	Thank you for joining this episode. This module is brought to you by **EduPyramids**. We are proud to be supported by **SINE**, **IIT Bombay**. We hope you learned something new today.